

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12

Mathematics Term 1 Examination

March 2005

TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page.
- At the end of the examination this question sheet must be attached to the front of your answers.
- Attempt all questions.
- Start all questions on a new page.
- All necessary working must be shown.

Q 1	Q 2	Q 3	Q 4	Q 5	TOTAL
/14	/15	/13	/13	/15	/70

QUESTION 1: (14 MARKS)

- (a) If the n th term of a sequence is given by $T_n = 3^{2-n}$
- 2 (i) show that the sequence is geometric
- 1 (ii) Find its common ratio
- 1 (iii) Write an expression for the sum to n terms.
- (b) Given the series $4 + 5\frac{1}{3} + 6\frac{2}{3} + \dots$
- 2 (i) find the 19th Term
- 2 (ii) find the sum to 19 terms
- 3 (c) If the limiting sum of the series $1 + x + x^2 + \dots$ is $\frac{3}{5}$ find the value of x .
- 3 (d) What is the value of the 11th term of the series

$$1\frac{1}{2} - 3 + 6 - \dots$$

QUESTION 2: (15 MARKS) (Start a new page)

- 2 (a) Find the value of $\sum_{n=1}^{10} 2^n$
- 2 (b) Find the slope of the tangent to the curve $y = x - \frac{1}{x}$ at the point where $x=1$
- 2 (c) By using calculus, explain why the curve $y = 5 + 2x - 3x^2$ is always concave down.
- 3 (d) Find the value of x for which the curve $y = x^2 - 5x + 3$ is increasing.
- 6 (e) If α and β are the roots of the equation $x^2 - 2x - 7 = 0$ find the value of:

(i) $\alpha + \beta$ (ii) $\alpha\beta$ (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ (iv) $\alpha^2 + \beta^2$

QUESTION 3: (13 Marks) (Start a new page)

- 3 (a) Form a quadratic equation containing no fractions which has roots of $-\frac{2}{3}$ and $\frac{3}{2}$
- 7 (b) (i) Find the maximum and minimum turning points on the curve $y = x^3 - 3x + 3$
- 3 (ii) Hence sketch the curve showing all essential features.
DO NOT find inflexion points or the x intercepts.

QUESTION 4: (13 MARKS) (Start a new page)

- 2 (a) Find the acute angle x (in degrees) which solves $\cos^2 x = \frac{1}{2}$
- 3 (b) Solve, to the nearest minute, the equation
- $$3 \sin \theta - 2 = 0 \quad \text{if} \quad 0^\circ \leq \theta \leq 360^\circ$$
- (c) The adjacent sides of a parallelogram have lengths of 4cm and 6cm and include an angle of 60°
- (i) Draw the parallelogram
- 3 (ii) Calculate the length of the longer diagonal, giving your answer correct to 2 dec. places.
- 2 (iii) Calculate the area of the parallelogram, correct to 2 dec. places.
- 3 (d) By making a suitable substitution, or otherwise, solve the equation:

$$x^4 + x^2 - 2 = 0$$

QUESTION 5 (15 Marks) (Start a new page)

- (a) If $y = ax^2 + bx + c$ passes through the point (0,7) and the curve has a turning point at (3,-2),

1

- (i) find the value of c

2

- (ii) by investigating $\frac{dy}{dx}$ prove that $6a + b = 0$

3

- (iii) by finding another equation in a and b , and using your answer to part (ii) above, find the values of a and b .

- (b) In the town of Mathsville, the population increases at a rate of 6% per year due to births and migration, but loses D persons per year due to deaths.

1

- (i) The population as at January 1, 2005 is 100 000. Find an expression for the population on January 1, 2006

2

- (ii) Show that the population on January 1, 2007 will be $112,360 - 2.06D$

3

- (iii) Show that the population after n years from January 1, 2005 is given by

$$P = 100\,000(1.06)^n - \frac{50D}{3}(1.06^n - 1)$$

3

- (iv) If the death rate (D) is a constant 500 persons per year, estimate the population as at January 1, 2025, to the nearest 100.

QUESTION 1:

(a) $T_n = 3^{2-n}$

(i) $T_1 = 3^1 = 3$

$T_2 = 3^0 = 1$

$T_3 = 3^{-1} = \frac{1}{3}$

$T_3/T_2 = T_2/T_1$

(2)

They may have to say that there is a common ratio of $\frac{1}{3}$ between terms (2)

(ii) $r = \frac{1}{3}$ (1)

(iii) $S_n = \frac{a(r^n - 1)}{r - 1}$

$= \frac{3(\frac{1}{3}^n - 1)}{\frac{1}{3} - 1} \leftarrow \text{this (or } \frac{3(1 - \frac{1}{3}^n)}{1 - \frac{1}{3}} \text{) is}$

sufficient for the marks.

(b) $4 + 5\frac{1}{3} + 6\frac{2}{3} +$

(i) $T_{19} = a + 18d$
 $= 4 + 18(1\frac{1}{3})$
 $= 28$

basically there are (2) marks for 28 or (1) for identifying $a = 4$ and $d = 1\frac{1}{3}$ and (1) for the formula $a + 18d$.

(ii) $S_{19} = \frac{19}{2} [4 + 28] \leftarrow (1) \text{ or } S_n = \frac{19}{2} [5 + 18(1\frac{1}{3})] (1)$
 $= 304 \leftarrow (1) \text{ for the answer}$

(c) $1 + x + x^2 + \dots = \frac{3}{5}$

$a = 1, r = x \quad S_\infty = \frac{1}{1-x} = \frac{3}{5} \leftarrow (1)$

$(1) \text{ or } 1 - 3x = 5$

$3x = -2$

$x = -\frac{2}{3} (1)$

(d) $a = \frac{1}{2}, r = -2$

$T_{10} = \frac{1}{2}(-2)^{10} \leftarrow (1) \text{ if this becomes } (-3)^{10} \text{ they}$

Question 2:

(a) $\sum_{i=1}^{10} 2^i = 2 + 4 + \dots + 2^{10} \leftarrow \textcircled{1}$
 $= \frac{2(2^{10}-1)}{2-1} \leftarrow \textcircled{1}$
 $= 2046$

(b) $\frac{dy}{dx} = 1 + \frac{1}{x^2}$ (or $1 + x^{-2}$) $\leftarrow \textcircled{1}$

At $x=1$ $m_T = 1 + \frac{1}{1}$
 $= 2 \leftarrow \textcircled{1}$

(c)

$$y = 5 + 2x - 3x^2$$

$$\frac{dy}{dx} = 2 - 6x$$

$$\frac{d^2y}{dx^2} = -6 \leftarrow \textcircled{1} \text{ for getting to here.}$$

always concave down because $\frac{d^2y}{dx^2} < 0 \leftarrow \textcircled{1}$ for a statement like this
(they must make this statement)

(d)

$$y = x^2 - 5x + 3$$

$$\frac{dy}{dx} = 2x - 5 \leftarrow \textcircled{1}$$

For increasing, $\frac{dy}{dx} > 0 \leftarrow \textcircled{1}$ for making this statement or showing they know they have to make $\frac{dy}{dx} > 0$

$$2x - 5 > 0$$

$$\therefore x > \frac{5}{2} \leftarrow \textcircled{1}$$

(e) $f(x) = x^2 - 2x - 7$

(i) $\alpha + \beta = 2 \leftarrow \textcircled{1}$

(ii) $\alpha\beta = -7 \leftarrow \textcircled{1}$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} \leftarrow \textcircled{1} \text{ for this step}$
 $= \frac{-2}{-7} \leftarrow \textcircled{1}$

QUESTION 3.

(a) (i) EITHER

OR $\alpha\beta = -1 \leftarrow \textcircled{1}$

$\alpha + \beta = \frac{5}{6} \leftarrow \textcircled{1}$

$(x + \frac{2}{3})(x - \frac{3}{2}) = 0 \leftarrow \textcircled{1}$

$\therefore (3x+2)(2x-3) = 0 \leftarrow \textcircled{2}$ OR $x^2 - \frac{5}{6}x + 1 = 0 \leftarrow \textcircled{1}$

$6x^2 - 5x + 6 = 0 \leftarrow \text{this line is NOT necessary}$

$6x^2 - 5x + 6 = 0 \leftarrow \text{if doing it this way the last line IS necessary}$

(b) (i) $y = x^3 - 3x + 3$

$\frac{dy}{dx} = 3x^2 - 3 \leftarrow \textcircled{1}$

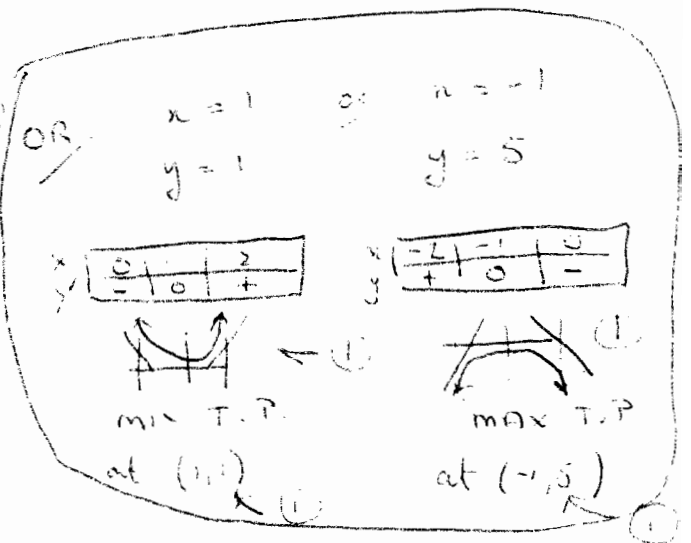
$\frac{d^2y}{dx^2} = 6x \leftarrow \textcircled{1}$

At S.P.s, $\frac{dy}{dx} = 0 \leftarrow \textcircled{1}$ for solving this

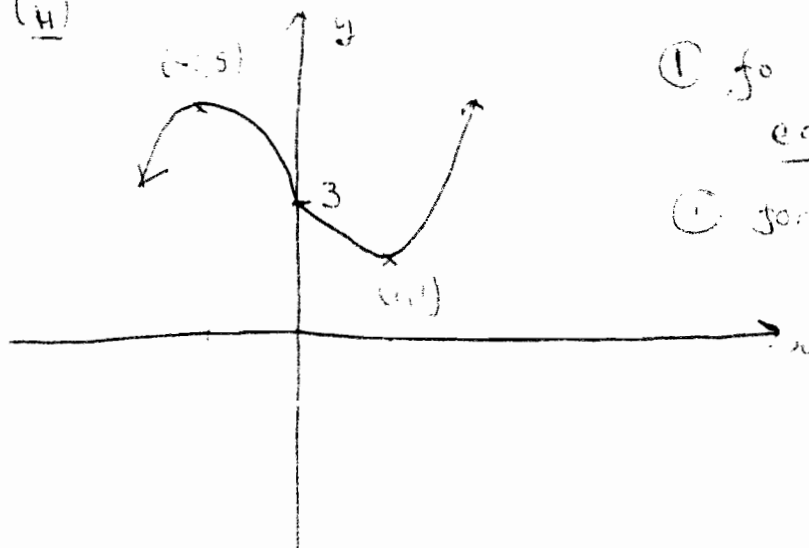
$\therefore 3x^2 - 3 = 0$

$\left\{ \begin{array}{l} x = 1 \\ y = 1 \\ y'' = 6 > 0 \end{array} \right\} \leftarrow \textcircled{1}$ min T.P. at (1,1) $\leftarrow \textcircled{1}$

$\left\{ \begin{array}{l} x = -1 \\ y = 5 \\ y'' = -6 < 0 \end{array} \right\} \leftarrow \textcircled{1}$ max T.P. at (-1,5) $\leftarrow \textcircled{1}$



(ii)



$\textcircled{1}$ for showing and labelling each T.P.

$\textcircled{1}$ for the point (0,3)

Question 4:

(a) $\cos^2 x = \frac{1}{2}$

$\cos x = \pm \frac{1}{\sqrt{2}}$

$\therefore x = 45^\circ$

award (2) for correct answer
award (1) for including 135°
award (1) for getting to step 2
only and not getting 45°

(b)

$3 \sin \theta - 2 = 0$

$3 \sin \theta = 2$

$\sin \theta = \frac{2}{3}$

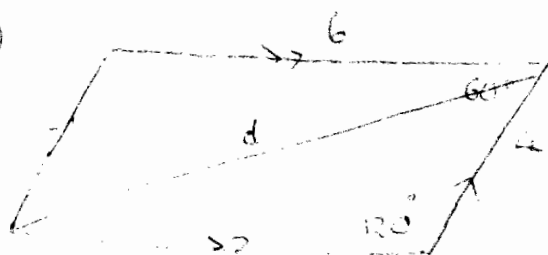
← (1)

$\therefore \theta =$

or

← (2) 1 for each answer

(c) (i)



no marks for this

(ii) $d^2 = 6^2 + 4^2 - 2 \cdot 6 \cdot 4 \cdot \cos 120^\circ$ ← (1) for this

$d = \sqrt{28}$

← (1)

← (1) for cos rule correct

(iii) $A = \frac{1}{2} \cdot 6 \cdot 4 \cdot \sin 120^\circ$ ← (1)
 $= 6\sqrt{3}$

$A =$

← (1) for double area is only 1 mark if only half the plgn

(d) EITHER

$(x^2 + 2)(x^2 - 1) = 0$ ← (1)

$x^2 = -2$ or $x^2 = 1$

no soln

$x = \pm 1$

← (2)

1 mark for the final answer

OR $x^2 + 2 = 0$

$x^2 + 2 = 0$

$(x + 2)(x - 1) = 0$ ← (1)

$x^2 = -2$ or $x^2 = 1$

no soln

$x = \pm 1$

←

QUESTION 5

(a) $y = ax^2 + bx + c$

(i) Passes through $(0, 7)$

$\therefore c = 7$ (1)

(ii) $\frac{dy}{dx} = 2ax + b$

Turning Point at $(3, -2)$

$\therefore 2a(3) + b = 0$ (1)

$6a + b = 0$ (1)

(iii) Passes through $(3, -2)$

$\therefore -2 = 9a + 3b + 7$ (1)

$\therefore 9a + 3b + 7 = 0$

$\therefore 3a + b + 3 = 0$ (1)

$6a + b = 0$ (1)

(b) (i) $3a + b + 3 = 0$

$a = 1$
 $b = -6$

2 marks

the process is important
only penalise one for
an incorrect sign, not
for having a and b both
wrong

(b) (i)

$P = 100,000(1.06) - D$ (1 mark - not required to
write $106,000 - D$)

(ii) $P = (100,000(1.06) - D)1.06 - D$
 $= 106,000(1.06) - D(1 + 1.06)$ (1)
 $= 112,360 - 2.06D$ (1)

(iii) $P_n = 100,000(1.06)^n - D(1 + 1.06 + \dots + 1.06^{n-1})$ (1)
 $S_n = \frac{1[(1.06)^n - 1]}{1.06 - 1}$ (1)
 $= 100,000(1.06)^n - \frac{500}{3}(1.06^n - 1)$ (1)

(iv) $P_{20} = 100,000(1.06)^{20} - \frac{25000}{3}(1.06^{20} - 1)$ { 1 mark for
 $n = 20$
1 mark for
expression
in }
 $= 320,713.5472 - 18392.796$
 $= 302,320.7512$